

Planetary Gearbox - 4-Stage Teardown and Repair Analysis

A. Job information

Customer name

Work order number

Date received

Manufacturer

Model

Serial number

Overall reduction

: 1

Stage 1 ratio

: 1

Stage 2 ratio

: 1

Stage 3 ratio

: 1

Stage 4 ratio

: 1

Oil capacity

Oil grade

Application / driven equipment

Service hours (if known)

Reported failure / customer complaint

B. As-received condition

☐

Photos taken as received

Notes

☐

External damage, cracks, missing parts

Notes

☐

Nameplate present and legible

Notes

Ram Gear Gearbox Evaluation

Shafts turn freely by hand: ☐ Yes ☐ No

Notes

☐ Noise or roughness when turned

Notes

☐ Oil leaks at seals / split line / plugs

Notes

☐ Breather condition

Notes

C. Oil condition

Oil drained - volume recovered

Oil sample ID (lab analysis)

☐ Oil sample taken for lab analysis

Appearance: ☐ Clean ☐ Dark ☐ Milky (water) ☐ Sludge ☐ Metallic particles

Magnetic drain plug debris description

D. As-found measurements before teardown

Endplay - Input shaft

Endplay - Output shaft

Output rotational backlash (input locked)

Ext. runout - Input shaft

Ext. runout - Output shaft

Ram Gear Gearbox Evaluation

E. Teardown checklist

- ☐ Match-mark housing, ring gears, carriers and shafts before disassembly
- ☐ Parts tagged and kept in order, by planetary stage
- ☐ Seals removed and inspected

As-found shim record

Shaft	Drive end (DE) as-found shim (in/mm)	Non-drive end (NDE) as-found shim (in/mm)	Shim pack	
Input			☐ Replace	☐ Reuse
Output			☐ Replace	☐ Reuse

P1. Planetary stage 1

Sun gear teeth (Z sun)	Ring gear teeth (Z ring)	Planet gear teeth	Number of planets
Stage ratio, calculated (1 + Z ring / Z sun)		Stage ratio, measured (input turns per carrier turn)	
: 1		: 1	

Stage 1 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling
☐ Spline wear / fretting ☐ Cracks / breakage

Ram Gear Gearbox Evaluation

Stage 1 sun gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet gears

Stage 1 planets (one row per planet)

Planet	Gear teeth condition	Pin / shaft wear & fit	Bearing / needle rollers	Thrust washers	Planet		
Planet 1					☐ Reuse	☐ Repair	☐ Replace
Planet 2					☐ Reuse	☐ Repair	☐ Replace
Planet 3					☐ Reuse	☐ Repair	☐ Replace
Planet 4					☐ Reuse	☐ Repair	☐ Replace
Planet 5					☐ Reuse	☐ Repair	☐ Replace
Planet 6					☐ Reuse	☐ Repair	☐ Replace

Stage 1 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 1 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 1 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

Ram Gear Gearbox Evaluation

Stage 1 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval
☐ Carrier bearing damage ☐ Spline / output coupling wear

Stage 1 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 1 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Stage 1 coupling / spline, stage 1 to stage 2

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

P2. Planetary stage 2

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated ($1 + Z \text{ ring} / Z \text{ sun}$)

 : 1

Stage ratio, measured (input turns per carrier turn)

 : 1

Stage 2 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Ram Gear Gearbox Evaluation

Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling
☐ Spline wear / fretting ☐ Cracks / breakage

Stage 2 sun gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet gears

Stage 2 planets (one row per planet)

Planet	Gear teeth condition	Pin / shaft wear & fit	Bearing / needle rollers	Thrust washers	Planet		
Planet 1					☐ Reuse	☐ Repair	☐ Replace
Planet 2					☐ Reuse	☐ Repair	☐ Replace
Planet 3					☐ Reuse	☐ Repair	☐ Replace
Planet 4					☐ Reuse	☐ Repair	☐ Replace
Planet 5					☐ Reuse	☐ Repair	☐ Replace
Planet 6					☐ Reuse	☐ Repair	☐ Replace

Stage 2 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 2 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 2 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks

Ram Gear Gearbox Evaluation

☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

Stage 2 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval

☐ Carrier bearing damage ☐ Spline / output coupling wear

Stage 2 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 2 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Stage 2 coupling / spline, stage 2 to stage 3

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

P3. Planetary stage 3

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated (1 + Z ring / Z sun)

 : 1

Stage ratio, measured (input turns per carrier turn)

 : 1

Stage 3 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐

Obtain all dimensional, endplay, backlash, and clearance specs from OEM drawings and bearing manufacturer data.

Ram Gear Gearbox Evaluation

Measurement	Measured	Spec	OK
Thrust washer thickness			☐

Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling
☐ Spline wear / fretting ☐ Cracks / breakage

Stage 3 sun gear	☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:	

Planet gears

Stage 3 planets (one row per planet)

Planet	Gear teeth condition	Pin / shaft wear & fit	Bearing / needle rollers	Thrust washers	Planet		
Planet 1					☐ Reuse	☐ Repair	☐ Replace
Planet 2					☐ Reuse	☐ Repair	☐ Replace
Planet 3					☐ Reuse	☐ Repair	☐ Replace
Planet 4					☐ Reuse	☐ Repair	☐ Replace
Planet 5					☐ Reuse	☐ Repair	☐ Replace
Planet 6					☐ Reuse	☐ Repair	☐ Replace

Stage 3 planet pins / shafts	☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:	

Stage 3 planet bearings / needle rollers	☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:	

Stage 3 thrust washers	☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:	

Ram Gear Gearbox Evaluation

Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

Stage 3 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval
☐ Carrier bearing damage ☐ Spline / output coupling wear

Stage 3 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 3 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Stage 3 coupling / spline, stage 3 to stage 4

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

P4. Planetary stage 4

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated (1 + Z ring / Z sun)

: 1

Stage ratio, measured (input turns per carrier turn)

: 1

Ram Gear Gearbox Evaluation

Stage 4 measurements

Measurement	Measured	Spec	OK
Sun-planet backlash			☐
Planet-ring backlash			☐
Sun gear float / axial play			☐
Carrier endplay			☐
Thrust washer thickness			☐

Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling
☐ Spline wear / fretting ☐ Cracks / breakage

Stage 4 sun gear		☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:		

Planet gears

Stage 4 planets (one row per planet)

Planet	Gear teeth condition	Pin / shaft wear & fit	Bearing / needle rollers	Thrust washers	Planet		
Planet 1					☐ Reuse	☐ Repair	☐ Replace
Planet 2					☐ Reuse	☐ Repair	☐ Replace
Planet 3					☐ Reuse	☐ Repair	☐ Replace
Planet 4					☐ Reuse	☐ Repair	☐ Replace
Planet 5					☐ Reuse	☐ Repair	☐ Replace
Planet 6					☐ Reuse	☐ Repair	☐ Replace

Ram Gear Gearbox Evaluation

Stage 4 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 4 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 4 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

Stage 4 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval
☐ Carrier bearing damage ☐ Spline / output coupling wear

Stage 4 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Stage 4 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

Stage 4 output spline / coupling (stage 4 carrier to output)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Ram Gear Gearbox Evaluation

F. Input, output, housing and lubrication

Input shaft

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Input seal

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Input shaft bearings

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Output shaft

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Output shaft bearings

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

Output seals

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Housing and bearing bores

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Lubrication: breather, plugs, sight glass, oil lines

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Bearing caps and shims

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Ram Gear Gearbox Evaluation

Other:		☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:		

Other:		☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:		

Other:		☐ Reuse	☐ Repair	☐ Replace
Part no.:		Findings:		

G. Failure modes observed

Gears: ☐ Pitting ☐ Spalling ☐ Scuffing/scoring ☐ Wear ☐ Tooth breakage
☐ Plastic deformation ☐ Corrosion ☐ Misaligned contact pattern

Bearings: ☐ Spalling/fatigue ☐ Brinelling ☐ Contamination dents
☐ Overheating/discoloration ☐ Corrosion/etching ☐ Electrical fluting
☐ Cage damage ☐ Skidding

Planetary: ☐ Planet pin wear ☐ Needle roller damage ☐ Thrust washer wear
☐ Carrier cracks ☐ Spline fretting ☐ Ring gear loose in housing

Location / notes

H. Dimensional checks

Bearing bore and journal diameters

Location	Bearing bore measured	Bearing bore spec	Bore OK	Journal measured	Journal spec	Journal OK
Input shaft bearing			☐			☐
Output shaft bearing			☐			☐
Stage 1 carrier bearing			☐			☐
Stage 2 carrier bearing			☐			☐
Stage 3 carrier bearing			☐			☐
Stage 4 carrier bearing			☐			☐

Seal journal condition notes

I. Root cause

- ☐ Lubrication failure ☐ Contamination ☐ Overload / shock load ☐ Misalignment
- ☐ Improper installation ☐ Fatigue / end of life ☐ Manufacturing defect
- ☐ Other

Root cause narrative

J. Repair summary

Parts to replace

Ram Gear Gearbox Evaluation

Machining / repair work required

Recommended upgrades or changes

Estimated labor hours

Estimated parts cost

Quote number

Recommendation:

☐

Repair

☐

Replace unit

☐

Not economical to repair

K. Sign-off

Inspected by

Name

Signature (type name)

Date

Reviewed / approved by

Name

Signature (type name)

Date

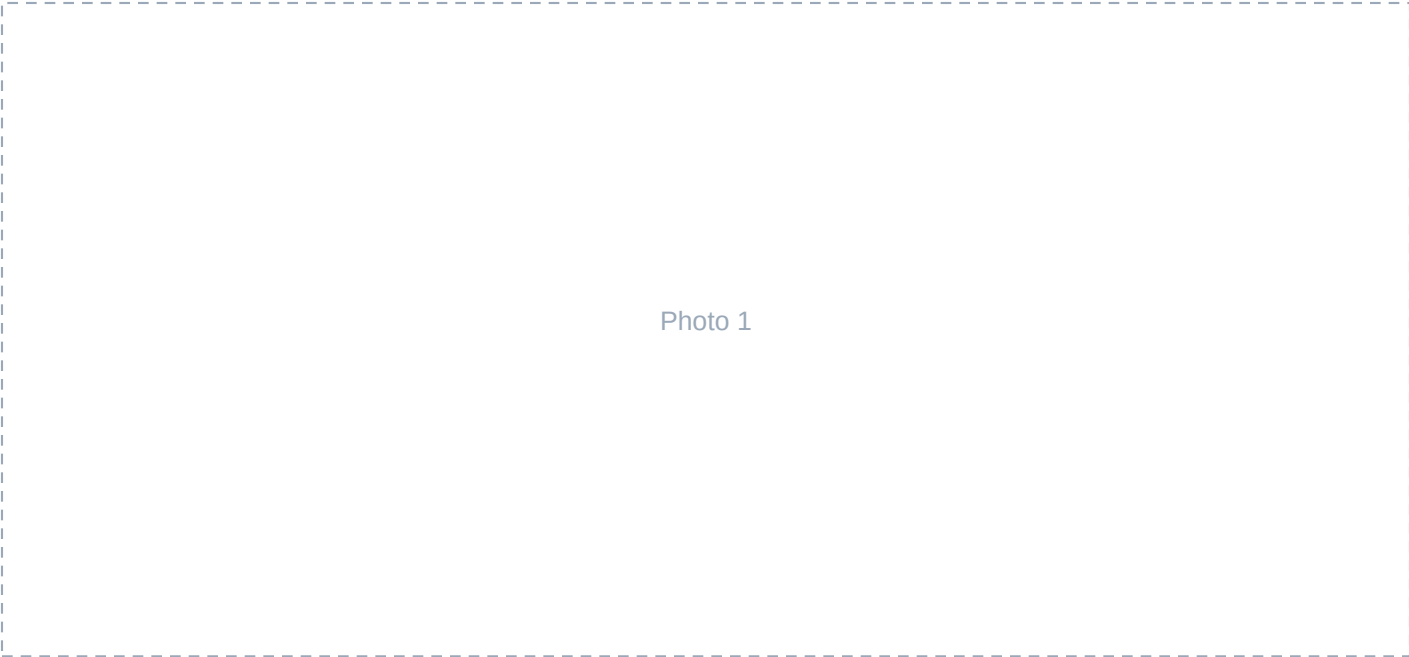
☐

Customer approval received

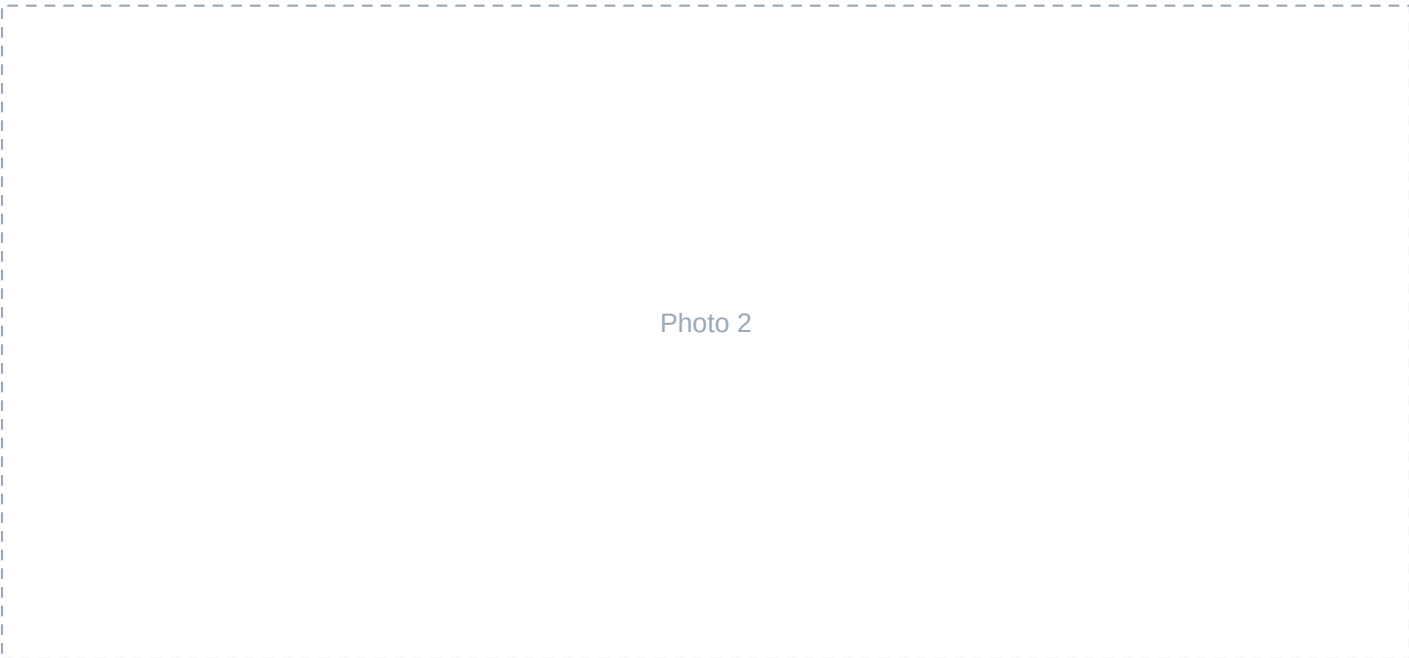
Customer approval date

L. Teardown photos

Attach teardown photos. Label each photo with a short caption, e.g. As received, Disassembled or a component name. Forms completed in the app print the photos and captions in this section.



Caption: _____



Caption: _____